



Tobias Stål

GEOSCIENTIST AND ENGINEER

Tasmania, Australia

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Geophysicist working on Antarctica's ice-covered interior. I integrate geology, geophysics and statistical modelling to understand interactions between the ice sheet and the solid Earth, with a focus on geothermal heat, subglacial geology and instrumentation. Background in geology and geomorphology from the Arctic and Europe, and an earlier professional career in sound, lighting and electrical engineering.

Academic positions

2022 – Present **Research Associate in Computational Physics (Geophysics)**, ACEAS, University of Tasmania, Tasmania, Australia

2021 – 2022 **Research Fellow in Antarctic Seismology**, University of Tasmania, Tasmania, Australia

Selected non-academic work experience

2021 – Present **GRIT Facility Manager – Instrumentation**, University of Tasmania, Tasmania, Australia. Instrumentation management and Antarctic fieldwork support

2010 – 2015 **Field Leader and Geophysicist**, COWI, Denmark. MEP, GPR and TEM fieldwork and survey design

1997 – 2016 **Sound engineer, sound, lighting and video designer, technical producer**, Swedish Radio, Royal Dramatic Theatre, Circus Cirkör, National Theatre of Ghana, STÜ and others, Sweden, Denmark, Estonia, Ghana, Palestine

Education

2021 **PhD, Geophysics, Geology and Computational Methods**, University of Tasmania, Australia

2015 **MSc, Geophysics**, University of Copenhagen, Denmark

2011 **BSc, Geology**, University of Copenhagen, Denmark

1998 **AD, Sound Engineering**, Piteå Music Conservatory (Luleå University of Technology), Piteå, Sweden

Selected publications

2025 NJ Abram, A Purich, ..., **T Stål** et al. — Emerging evidence of abrupt changes in the Antarctic environment. *Nature*.

2024 **T Stål**, JA Halpin, JW Goodge et al. — Geology Matters for Antarctic Geothermal Heat. *Geophysical Research Letters*.

2022 **T Stål**, AM Reading, S Fuchs et al. — Properties and biases of the global heat flow compilation. *Frontiers in Earth Science*.

2022 AM Reading, **T Stål**, JA Halpin et al. — Antarctic geothermal heat flow and its implications for tectonics and ice sheets. *Nature Reviews Earth & Environment*.

2021 **T Stål**, AM Reading, JA Halpin et al. — Antarctic Geothermal Heat Flow Model: Aq1. *Geochemistry, Geophysics, Geosystems*.

2020 **T Stål**, AM Reading, JA Halpin et al. — The Antarctic Crust and Upper Mantle: A Flexible 3D Model and Software Framework for Interdisciplinary Research. *Frontiers in Earth Science*.

2019 **T Stål**, AM Reading, JA Halpin et al. — A Multivariate Approach for Mapping Lithospheric Domain Boundaries in East Antarctica. *Geophysical Research Letters*.

Datasets

2021 **T Stål**, AM Reading, S Kupis et al. — Casey-Wilkins Adaptable Array [Dataset]. *International Federation of Digital Seismograph Networks*.

Awards and recognition

2023 Royal Society of Tasmania Doctoral Award

2021 Wiley Top Cited Article

2020 PhD Award for Outstanding Performance during Postgraduate Studies

Memberships and service

2023 – Present Member, International Heat Flow Commission (IHFC), IUGG

2021 – Present Co-chair, SCAR INSTANT Heat Flow Subcommittee

Art collaborations (as a scientist)

2025 T Stål, AM Reading, K Verell — *Seismic Dance Party*, Beaker Street (Hobart). Denman Glacier seismic data as an audio-visual dancefloor experience.

2025 L Bleach — *An Infrasonorous Archipelago*, Walk&Talk Biennial "Gestures of Abundance" (Azores). Seismo-volcanic infrasound as tactile and sonic works.

Other skills and experience

Coding Python (scikit-learn, TensorFlow, dask, xarray, geopandas), GIS (QGIS, ArcGIS, GDAL), LaTeX/TikZ, Bash, git, SCons, SQL, Seismic Unix, Madagascar, Blender

Certificates Antarctic trip leader and terrain-vehicle driver (AAD, Casey 2023); Wilderness First Aid; Coastal Skipper; PADI Advanced Open Water

Languages English (C2), Swedish (native), Danish (fluent), French (intermediate), Russian and Asante Twi (functional)